CIS/CS 445 Database Management

Course Project Requirements

and Rubric

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CIS/CS 445 Database Management

Overview

You will be designing a database for your Course Project from the ground up for a fictional *Regis University Accommodation Office*. In the project, you will perform the database life cycle of database analysis, database design, and implementation using the Oracle RDBMS.

Throughout this course, you will be working on seven Course Project Components described in the following pages using the Scenario provided below. Using the Oracle environment you set up in Week 1, you will be creating a database and performing queries against your database for your Course Project throughout this course. These requirements and rubric are located in the Course Resources folder for easy access throughout the course.

You have two submissions for your Course Project:

1. Your first submission will be in Week 5 and includes Course Project Components 1 and 2 (Dataset Definitions, and normalized ERD).
2. Your second submission will be in Week 8 and includes all seven of the Course Project Components, which include your revised Components 1 and 2.

Course Project Q & A Discussion Forum

Throughout the course, use the Course Project Q & A Discussion Forum to ask questions you may have. Your facilitator will respond in the forum unless the response is private in nature.

Scenario - Regis University Accommodation Office (Student Housing)

The director of the *Regis University Accommodation Office* requires you to design a database to assist with the administration of the office and the renting of residences to students. The requirements collection and analysis phase of the database design process has provided the following data requirements specification for the *Regis University Accommodation Office* database.

Data Requirements

The housing data stored for each full-time student includes: the student ID number, name (first and last name), home address (street, city, zip code), mobile phone number, email, date of birth, gender, category of student (for example, first-year undergraduate, postgraduate), nationality, special needs, any additional comments, current status (placed/waiting), major, and minor.

The student information stored relates to those currently renting a room and those on the waiting list. Students rent a room in a hall of residence.

When a student joins Regis University, he or she is assigned to a member of staff who acts as his or her Advisor. The Advisor is responsible for monitoring the student's welfare and academic progression throughout his or her time at Regis University. The data held on a student's Advisor includes full name, position, department name, internal telephone number, email, and room number.

Halls of Residence

Each hall of residence has a name, address, telephone number, and a hall manager, who supervises the operation of the hall. The halls provide only single rooms, which have a room number, place number, and monthly rent rate.

The place number uniquely identifies each room in all halls controlled by the Accommodation Office and is used when renting a room to a student.

Leases

A student may rent a room in a hall for various periods of time. New lease agreements are negotiated at the start of each academic year, with a minimum rental period of one semester and a maximum rental period of one year, which includes three semesters: one, two and the summer semester. Each individual lease agreement between a student and the Accommodation Office is uniquely identified using a lease number.

The data stored on each lease includes the lease number, duration of the lease (given as semesters), student's name and ID number, place number, room number, address details of the hall of residence, and the date the student wishes to enter the room, and the date the student wishes to leave the room (if known).

Invoices

At the start of each semester, each student is sent an invoice for the following rental period. Each invoice has a unique invoice number.

The data stored on each invoice includes the invoice number, lease number, semester, payment due, student's full name and ID number, place number, room number, and the address of the hall. Additional data is also held regarding the payment of the invoice and includes the date the invoice was paid, the method of payment (check, cash, Visa, and so on), and the date the first and second reminder was sent (if necessary).

Halls of Residence inspections

Halls of residence rooms are inspected by staff on a regular basis to ensure that the accommodation is well maintained. The information recorded for each inspection is the name of the member of staff who carried out the inspection, the date of inspection, an indication of whether the room was found to be in a satisfactory condition (yes or no), and any additional comments.

Residence staff

Some information is also held on members of staff of the Accommodation Office and includes the staff number, name (first and last name), email, home address (street, city, postcode), date of birth, gender, position (for example, Hall Manager, Administrative Assistant, Cleaner) and location (for example, Accommodation Office or Hall).

Courses

The Accommodation Office also stores a limited amount of information on the courses offered by the university, including the course number, course title (including year), course instructor, instructor's on-campus telephone number, email, room number, and department name. Each student is also associated with a single program of study.

Next-of-kin

Whenever possible, information on a student's next-of-kin is stored, which includes the name, relationship, address (street, city, postcode), and contact telephone number.

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Course Project Components/Student Deliverables

The deliverables for your Course Project are:

Component 1: Datasets for the *Regis University Accommodation Office.* This is a list of all the datasets and data attributes that the Office needs to function. For example, a STUDENT dataset containing StudentIDNumber, StudentFirstName, etc.

Component 2: Normalized Entity-Relationship Diagram (ERD) to 3NF for the *Regis University Accommodation Office* using Crow's Feet notation. Please show all data modeling work which includes entities, entity attributes, cardinality of relationships, key constraints, and relevant assumptions made in modeling the data. Must use a data modeling tool such as MS Visio or Oracle SQL Developer Data Modeler.

Component 3: Data Dictionary with all of your attributes using your very detailed ER diagram. This should include the table names, field names, data types, field lengths, constraints, associated relations, and field name descriptions.

Component 4: SQL Script (.sql file) to create your database tables in Oracle (using SQL Plus, SQL Developer or Data Modeler)

Component 5: SQL Script (.sql file) to populate your database tables with at least three records (using SQL Plus, SQL Developer or Data Modeler).

Component 6: SQL Script (.sql file) with the following query transactions against your database (using SQL Plus, SQL Developer or Data Modeler):

a. Present a report listing the Manager's name and telephone number for each hall of residence.

b. Present a report listing the names and ID numbers of students with the details of their lease agreements.

c. Display the total number of students in each student category.

d. Display the details of housing inspections where the property was found to be in unsatisfactory condition.

e. Present a report of the names and student IDs with their room number and hall of residence. Group by hall of residence.

Component 7: Output from each of the query transactions in item 6 above. You will need to cut and paste from Oracle's SQL Plus, SQL Developer or Data Modeler into a MS Word .doc file. You can do this by selecting the query output in SQL PLUS, pressing <ctrl><alt>PrtScn keys or use the Windows Snipping Tool, and then pasting into the Word document file.

Schedule of Work

Week	Course Project - Components	Student Deliverables
2	Review Course Project Requirements & Rubric	None

	(located in Course Resources folder for easy access throughout this course)	
3	Component 1 – Dataset Definitions (see details below)	None
4	Component 2 - Data Modeling and Analysis	None
5	Component 3 - Data Dictionary	Submit Components 1 & 2 (only) to Dropbox
6	Components 4 & 5 – DDL & DML	None
7	Component 6 - Database Queries	None
8	Component 7 – Project Format	Submit all seven Components to Dropbox; includes revised Components 1 & 2

Submission

Create a single zipped folder that contains all seven Course Project Components which includes documents and any SQL script files used in your project. Include a summary with any assumptions you have made.

By Sunday of Week 8, submit your zipped Course Project to the Dropbox titled for this activity.

Evaluation Criteria

See the *Course Project Rubric* below for more information about how you will be evaluated for each Component.

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Course Project Rubric

Student Name: _____ Term: _____ (i.e. 15S8W1)

Date: _____ Facilitator: _____

Criteria	*Exemplary*	*Proficient*	*Basic*	*Poor*	*Points Earned*
TECHNOLOGY COMPONENT					
Project Component 1: Dataset Definitions Identifies datasets necessary for the project's database.	Identifies and clearly defines **all** of the datasets and their attributes needed for the project.	Identifies and clearly defines **most** of the datasets and their attributes needed for the project.	Identifies and clearly defines **some** of the datasets and their attributes needed for the project.	Identifies and clearly defines **few** or **none** of the datasets and their attributes needed for the project.	
Project Component 1: Points	10 - 9	8 - 7	6 - 3	2 - 0	
Project Component 2: Data Modeling and Analysis Identifies requirements for the project's database; determines the entities and relationships for the data; and constructs a key-based Entity Relationship Diagram (ERD).	Identifies and clearly documents **all** the entities needed for the project; illustrates **all** relationships with multiplicities between entities; illustrates **all** Primary and Foreign Key relationships; and produces a complete ERD of the proposed system.	Identifies and clearly documents **most** of the entities needed for the project; illustrates **most** of the relationships with multiplicities between entities; illustrates **most** of the Primary and Foreign Key relationships; and produces a near-complete ERD of the proposed system.	Identifies and documents **some** of the entities needed for the project; illustrates **some** of the relationships with multiplicities between entities; illustrates **some** of the Primary and Foreign Key relationships; and produces an incomplete ERD of the proposed system.	Identifies and documents **few** of the entities needed for the project; illustrates **few** of the relationships with multiplicities between entities; illustrates **few** of the Primary and Foreign Key relationships; and produces an incomplete ERD of the proposed system.	
Project Component 2: Points	20 - 18	17 - 14	13 - 6	5 - 0	

Project Component 3: Data Dictionary Provide a Data Dictionary for the project which provides a detailed accounting of all tables found in the database.	Describes **all** tables in the database. Within each table defines **all** of the following**:** attribute name, attribute description, data type, data format, field length, constraints, and primary/foreign key relationships.	Describes **most** tables in the database. Within each table defines **most** of the following: attribute name, attribute description, data type, data format, field length, constraints, and primary/foreign key relationships.	Describes **some** tables in the database. Within each table defines **some** of the following: attribute name, attribute description, data type, data format, field length, constraints, and primary/foreign key relationships.	Describes **few or no** tables in the database. Within each table defines **few or none** of the following: attribute name, attribute description, data type, data format, field length, constraints, and primary/foreign key relationships.	
Project Component 3: Points	10 - 9	8 - 7	6 - 3	2 - 0	
Project Component 4: Data Definition Language (DDL) Construct the necessary DDL to create the relational database for the project using the Data Model and Data Dictionary.	Constructs the DDL necessary to create **all** of the relational database tables for the project. Illustrates **successful** execution of DDL.	Constructs the DDL necessary to create **most** of the relational database tables for the project. Illustrates **successful** execution of DDL.	Constructs the DDL necessary to create **some** of the relational database tables for the project. **May or may not** illustrate successful execution of DDL.	Constructs the DDL necessary to create **few or none** of the relational database tables for the project. **Does not** illustrate successful execution of DDL.	
Project Component 4: Points	10 - 9	8 - 7	6 - 3	2 - 0	
Project Component 5: Data Manipulation Language (DML) Construct the necessary DML to populate the relational database for the project.	Constructs the DML necessary to populate **all** of the relational database tables for the project with three data records. Illustrates **successful** execution of DML.	Constructs the DML necessary to populate **most** of the relational database tables for the project with three data records. Illustrates **successful** execution of	Constructs the DML necessary to populate **some** of the relational database tables for the project with three records of data. **May or may not**	Constructs the DML necessary to populate **few or none** of the relational database tables for the project with three records of data. **Does not** illustrate	

		DML.	illustrate successful execution of DML.	successful execution of DML.	
Project Component 5: Points	10 - 9	8 - 7	6 - 3	2 - 0	
Project Component 6: Database Queries Construct the necessary DML to query the relational database for the required queries. Illustrates the successful execution of each query. Each query is worth 10 pts.	Constructs the DML necessary to query the relational database for **all three** queries. Illustrates **successful** execution of **all three** queries.	Constructs the DML necessary to query the relational database for **two** queries. Illustrates **successful** execution of at least **two** queries.	Constructs the DML necessary to query the relational database for **one** query. Illustrates **successful** execution of at least **one** query or **some attempt** at query output.	**Does not** construct the DML necessary to query the relational database or illustrate **successful** execution of **any** queries or **some attempt** at query output.	
Project Component 6: Points	30 - 26	25 - 21	20 – 9	8 - 0	
Project Component 7: Project Format Single Project document containing all seven project components.	Project document contains **all** required components.	Project document contains **most** required components.	Project document contains **some** required components.	Project is **not** in a single document.	
Project Component 7: Points	10 - 9	8 - 7	6 - 3	2 - 0	
TOTAL POINTS EARNED (out of 100 points)					
Facilitator Comments:					